

CITS5508 Practice Exam — Chapters 1-2

The ML Landscape & End-to-End ML Projects

- This practice paper has **5 questions** for **60 marks**.
 - Allow approximately **two minutes per mark** ($\approx$ 2 hours total).
 - Only UWA-approved calculators permitted.
 - Answers are in the separate file *Practice Exam - Chapters 1-2 (Answer Key).md*.
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Question 1. (12 marks)

(a) Tom Mitchell defined machine learning in terms of a task **T**, a performance measure **P**, and experience **E**. For a system that learns to recommend films to a streaming-service user, state what T, E and P could each be. (3 marks)

(b) Explain the difference between **supervised** and **unsupervised** learning. Give one real-world application of each, and in each case describe the **features** and (if any) the **response variable**. (4 marks)

(c) For each of the following, state whether it is best framed as supervised, unsupervised, semi-supervised, self-supervised, or reinforcement learning, and justify in one sentence: (3 marks) 1. Training a robot to walk over unknown terrain by trial and reward. 2. Grouping a website's visitors into segments with no predefined categories. 3. Pre-training an image model by masking patches and predicting the missing pixels.

(d) What is the difference between **instance-based** and **model-based** learning? Name one algorithm of each type. (2 marks)

Question 2. (12 marks)

(a) Define a model **parameter** and a model **hyperparameter**, and clearly state what distinguishes them. (3 marks)

(b) Explain what **overfitting** and **underfitting** mean. For each, give **two** possible remedies. (5 marks)

(c) Explain what **regularisation** is and how it relates to a model's **degrees of freedom**. Why does setting a regularisation hyperparameter to a very large value risk underfitting? (4 marks)

Question 3. (12 marks)

(a) You are choosing between three candidate models. Explain why you should **not** select the best one by repeatedly evaluating all three on the **test set**, and describe how a **validation set** (or cross-validation) avoids this problem. (4 marks)

(b) Describe **k-fold cross-validation**, and state one advantage and one disadvantage compared with a single holdout validation set. (4 marks)

(c) A team trains a flower-classifier on millions of images scraped from the web, but the app will run on photos taken with phone cameras. After training, validation accuracy is poor. Explain how a **train-dev set** would let them determine whether the problem is **overfitting** or **data mismatch**. (4 marks)

Question 4. (12 marks)

(a) A regression model produced the following predictions on five test instances:

Instance	Actual y_i	Predicted $\hat{y}_i$
1	10	12
2	20	18
3	30	35
4	40	39
5	50	38

Compute the **MAE** and the **RMSE** of these predictions. Show your working. (5 marks)

(b) Which of the two metrics is **more sensitive to outliers**, and why? With which norm (ℓ_1 or ℓ_2) is each associated? (3 marks)

(c) Explain **data-snooping bias** and the practice that prevents it. (2 marks)

(d) Explain what **stratified sampling** is and why it can give a better train/test split than purely random sampling. (2 marks)

Question 5. (12 marks)

(a) The following code prepares a dataset:

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Explain why `fit_transform` is called on the training set but only `transform` on the test set. What problem would arise if you called `fit_transform` on the test set as well? (4 marks)

(b) A dataset has a numeric feature with some missing values and a nominal categorical feature. Describe an appropriate preprocessing step for **each**, and explain why an `OrdinalEncoder` would be a poor choice for the categorical feature. (4 marks)

(c) Write a short scikit-learn snippet that builds a `Pipeline` (or `make_pipeline`) which median-imputes missing values, standardises the features, and finishes with a `LinearRegression` estimator. Briefly state one benefit of bundling these steps into a single pipeline. (4 marks)